

(a)

Inputs		Output
A	*B*	*Y*=*A*.*B*
0	0	0
0	1	0
1	0	0
1	1	1

(b)

Figure 10.41 (a) Two input AND gate (b) Truth table

Boolean equation:

$$Y = A.B$$

It performs logical multiplication and is different from arithmetic multiplication.

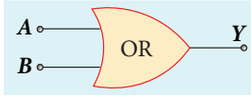
Logic operation

The output of AND gate is high only when all the inputs are high. In the rest of the cases, the output is low. It is represented in the truth table (Figure 10.41(b)).

OR gate

Circuit Symbol

The circuit symbol of a two input OR gate is shown in Figure 10.42(a). *A* and *B* are inputs and *Y* is the output.



(a)

Inputs		Output
A	*B*	*Y*= *A*+*B*
0	0	0
0	1	1
1	0	1
1	1	1

(b)

Figure 10.42 (a) Two input OR gate (b) Truth table

Boolean equation:

$$Y = A + B$$

It performs logical addition and is different from arithmetic addition.

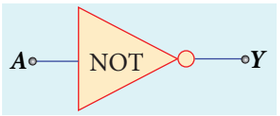
Logic operation

The output of OR gate is high (logic 1 state) when either of the inputs or both are high. The truth table of OR gate is shown in Figure 10.42(b).

NOT gate

Circuit symbol

The circuit symbol of NOT gate is shown in Figure 10.43(a). *A* is the input and *Y* is the output.



(a)

Inputs	Output
A	$Y = \overline{A}$
0	1
1	0

(b)

Figure 10.43 (a) NOT gate (b) Truth table

Boolean equation:

$$Y = \overline{A}$$

Logic operation

The output is the complement of the input. It is represented with an overbar. It is also called as inverter. The truth table infers that the output *Y* is 1 when input *A* is 0 and vice versa. The truth table of NOT is shown in Figure 10.43(b).

NAND gate

Circuit symbol

The circuit symbol of NAND gate is shown in Figure 10.44(a). *A* and *B* are inputs and *Y* is the output.

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